T3 matched EFT Lagrangian terms: EFT_2_after_S1_S2

EFT order: 5; loop order: 1. Rows are model configurations and columns are field configurations. This stage integrates out S1, S2. Remaining active T3 fields: none.

Notation key

symbol	interaction / definition
g_1	$D_\mu \supset i g_1 Y B_\mu$
g_2	$D_\mu \supset i g_2 T^A W_\mu^A$
g_3	$D_\mu \supset i g_3 t^A G_\mu^A$
Y_u	$-\bar{Q} Y_u \widetilde{H} u + \text{h.c.}$
Y_d	$-\bar{Q} Y_d H d + \text{h.c.}$
Y_e	$-\bar{L} Y_e H e + \text{h.c.}$
y_1	$L F S_1 + \text{h.c.}$
y_2	$L F S_2 + \text{h.c.}$
M_F	$-\bar{F} M_F F$
T	$\mathrm{Tr}(Y_e^\dagger Y_e + 3 Y_u^\dagger Y_u + 3 Y_d^\dagger Y_d)$
$T_\nu^{(1)}$	$\mathrm{Tr}(y_1^\dagger y_1)$
$T_\nu^{(2)}$	$\mathrm{Tr}(y_2^\dagger y_2)$
T_ν	$T_\nu^{(1)} + T_\nu^{(2)}$
m_1^2	$m_1^2 S_1^\dagger S_1$
m_2^2	$m_2^2 S_2^\dagger S_2$
λ_1	$\frac{1}{2}\lambda_1 (H^\dagger H)^2$
$\lambda_{S_1}^{(1)}$	$\frac{1}{2}\lambda_{S_1}^{(1)} (S_1^\dagger S_1)^2$
$\lambda_{S_2}^{(1)}$	$\frac{1}{2}\lambda_{S_2}^{(1)} (S_2^\dagger S_2)^2$
$\lambda_{HS_1}^{(1)}$	$\lambda_{HS_1}^{(1)} (H^\dagger H)(S_1^\dagger S_1)$
$\lambda_{HS_2}^{(1)}$	$\lambda_{HS_2}^{(1)} (H^\dagger H)(S_2^\dagger S_2)$
$\lambda_{HS_1}^{(A)}$	$\lambda_{HS_1}^{(A)} (H^\dagger T^A H)(S_1^\dagger T^A S_1)$
$\lambda_{HS_2}^{(A)}$	$\lambda_{HS_2}^{(A)} (H^\dagger T^A H)(S_2^\dagger T^A S_2)$
λ_5	$\lambda_5 H H S_1 S_2^\dagger + \text{h.c.}$
$\lambda_{12}^{(1)}$	$\lambda_{12}^{(1)} (S_1^\dagger S_1)(S_2^\dagger S_2)$
$\lambda_{12}^{(A)}$	$\lambda_{12}^{(A)} (S_1^\dagger T^A S_1)(S_2^\dagger T^A S_2)$
$\lambda_{12}^{(\times)}$	crossed independent S_1–S_2 contraction
$\lambda_{S_1}^{(A)}$	additional independent S_1 self-contraction
$\lambda_{S_2}^{(A)}$	additional independent S_2 self-contraction
α	$Y_{S_1} = \alpha/2,\ Y_F = (\alpha+1)/2,\ Y_{S_2} = (\alpha+2)/2$
$C_2(d)$	$(d^2-1)/4$

$T(d)$	$d(d^2-1)/12$
$C^{ij;AB}_{LLS_1S_2}$	$L_iL_jS_1^AS_2^B$
C_{ijab}	$L_iL_j\phi_a\phi_b$
C_5^{ij}	$(L_i^TCL_j)HH$
R_W	$16\pi^2\beta_{C_5} \supset R_W\lambda_5 C_{LLS_1S_2}$

Field combinations 1 of 2

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$H\,\ell$	$H\,\ell\,e$	$F\,H\,\ell$
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3}\lambda_5\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}\epsilon M_F}$ $+\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3}\lambda_5\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}M_F}$ $+\frac{2\hbar\overline{\lambda_5 L_{a_3,i_1}}\,C P_R\,\overline{L_{a_4,i_2}}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\sqrt{3}\epsilon M_F}$	$\frac{\hbar\overline{y_{1,i_3,r_1}}\,\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\hbar\overline{y_{1,i_3,r_1}}\,\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{\sqrt{3}\hbar\overline{y_{2,i_3,r_1}}\,\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\hbar\overline{y_{2,i_3,r_1}}\,\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$	$+\frac{2\hbar\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}{}^T C P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_5\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}M_F}$ $+\frac{2\hbar\log\left(\frac{\mu^2}{M_F^2}\right)\overline{\lambda_5 L_{a_3,i_1}}\,C P_R\,\overline{L_{a_4,i_2}}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\sqrt{3}M_F}$
cont.								$+\frac{2\hbar\overline{\lambda_5 L_{a_3,i_1}}\,C P_R\,\overline{L_{a_4,i_2}}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\sqrt{3}M_F}$	$+\frac{\hbar\overline{y_{1,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}{}^\dagger}{4\epsilon}$ $+\frac{3}{8}\hbar\overline{y_{1,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}{}^\dagger$ $+\frac{\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}{}^\dagger}{4\epsilon}$	
cont.									$+\frac{3}{8}\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}{}^\dagger$	
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3}\lambda_5\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}\epsilon M_F}$ $+\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3}\lambda_5\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}M_F}$ $+\frac{2\hbar\overline{\lambda_5 L_{a_3,i_1}}\,C P_R\,\overline{L_{a_4,i_2}}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\sqrt{3}\epsilon M_F}$	$\frac{\hbar\overline{y_{1,i_3,r_1}}\,\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\hbar\overline{y_{1,i_3,r_1}}\,\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{\sqrt{3}\hbar\overline{y_{2,i_3,r_1}}\,\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\hbar\overline{y_{2,i_3,r_1}}\,\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$	$+\frac{2\hbar\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}{}^T C P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_5\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}M_F}$ $+\frac{2\hbar\log\left(\frac{\mu^2}{M_F^2}\right)\overline{\lambda_5 L_{a_3,i_1}}\,C P_R\,\overline{L_{a_4,i_2}}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\sqrt{3}M_F}$
cont.								$+\frac{2\hbar\overline{\lambda_5 L_{a_3,i_1}}\,C P_R\,\overline{L_{a_4,i_2}}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\sqrt{3}M_F}$	$+\frac{\hbar\overline{y_{1,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}{}^\dagger}{4\epsilon}$ $+\frac{3}{8}\hbar\overline{y_{1,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}{}^\dagger$ $+\frac{\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}{}^\dagger}{4\epsilon}$	
cont.									$+\frac{3}{8}\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}{}^\dagger$	
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3}\lambda_5\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}\epsilon M_F}$ $+\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3}\lambda_5\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}M_F}$ $+\frac{2\hbar\overline{\lambda_5 L_{a_3,i_1}}\,C P_R\,\overline{L_{a_4,i_2}}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\sqrt{3}\epsilon M_F}$	$\frac{\hbar\overline{y_{1,i_3,r_1}}\,\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\hbar\overline{y_{1,i_3,r_1}}\,\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{\sqrt{3}\hbar\overline{y_{2,i_3,r_1}}\,\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\hbar\overline{y_{2,i_3,r_1}}\,\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$	$+\frac{2\hbar\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}{}^T C P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_5\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}M_F}$ $+\frac{2\hbar\log\left(\frac{\mu^2}{M_F^2}\right)\overline{\lambda_5 L_{a_3,i_1}}\,C P_R\,\overline{L_{a_4,i_2}}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\sqrt{3}M_F}$
cont.								$+\frac{2\hbar\overline{\lambda_5 L_{a_3,i_1}}\,C P_R\,\overline{L_{a_4,i_2}}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\sqrt{3}M_F}$	$+\frac{\hbar\overline{y_{1,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}{}^\dagger}{4\epsilon}$ $+\frac{3}{8}\hbar\overline{y_{1,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}{}^\dagger$ $+\frac{\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}{}^\dagger}{4\epsilon}$	
cont.									$+\frac{3}{8}\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}{}^\dagger$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$H\,\ell$	$H\,\ell\,e$	$F\,H\,\ell$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3}\lambda_5\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}\epsilon M_F}$ $+\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3}\lambda_5\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}M_F}$ $+\frac{2\hbar\overline{\lambda_5 L_{a_3,i_1}}\, C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4} H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\sqrt{3}\epsilon M_F}$	$\frac{\hbar\overline{y_{1,i_3,r_1}}\,\overline{L_{a_1,i_2}}\, P_R\, e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\hbar\overline{y_{1,i_3,r_1}}\,\overline{L_{a_1,i_2}}\, P_R\, e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}$ $+\frac{\sqrt{3}\hbar\overline{y_{2,i_3,r_1}}\,\overline{L_{a_1,i_2}}\, P_R\, e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\hbar\overline{y_{2,i_3,r_1}}\,\overline{L_{a_1,i_2}}\, P_R\, e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}$	$+\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3}\lambda_5\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}M_F}$ $+\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{\lambda_5 L_{a_3,i_1}}\, C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4} H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\sqrt{3}M_F}$
cont.								$+\frac{2\hbar\overline{\lambda_5 L_{a_3,i_1}}\, C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4} H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\sqrt{3}M_F}$	$+\frac{\hbar\overline{y_{1,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\, P_L\, L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}{}^\dagger}{4\epsilon}$ $+\frac{3}{8}\hbar\overline{y_{1,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\, P_L\, L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}{}^\dagger$ $+\frac{\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\, P_L\, L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}{}^\dagger}{4\epsilon}$	
cont.									$+\frac{3}{8}\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\, P_L\, L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}{}^\dagger$	
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$+\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3}\lambda_5\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}\epsilon M_F}$ $+\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3}\lambda_5\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}M_F}$ $+\frac{2\hbar\overline{\lambda_5 L_{a_3,i_1}}\, C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4} H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\sqrt{3}\epsilon M_F}$	$\frac{\hbar\overline{y_{1,i_3,r_1}}\,\overline{L_{a_1,i_2}}\, P_R\, e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\hbar\overline{y_{1,i_3,r_1}}\,\overline{L_{a_1,i_2}}\, P_R\, e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}$ $+\frac{\sqrt{3}\hbar\overline{y_{2,i_3,r_1}}\,\overline{L_{a_1,i_2}}\, P_R\, e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\hbar\overline{y_{2,i_3,r_1}}\,\overline{L_{a_1,i_2}}\, P_R\, e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}$	$+\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3}\lambda_5\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}M_F}$ $+\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{\lambda_5 L_{a_3,i_1}}\, C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4} H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\sqrt{3}M_F}$
cont.								$+\frac{2\hbar\overline{\lambda_5 L_{a_3,i_1}}\, C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4} H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\sqrt{3}M_F}$	$+\frac{\hbar\overline{y_{1,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\, P_L\, L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}{}^\dagger}{4\epsilon}$ $+\frac{3}{8}\hbar\overline{y_{1,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\, P_L\, L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}{}^\dagger$ $+\frac{\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\, P_L\, L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}{}^\dagger}{4\epsilon}$	
cont.									$+\frac{3}{8}\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\, P_L\, L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}{}^\dagger$	
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$+\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3}\lambda_5\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}\epsilon M_F}$ $+\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3}\lambda_5\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}M_F}$ $+\frac{2\hbar\overline{\lambda_5 L_{a_3,i_1}}\, C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4} H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\sqrt{3}\epsilon M_F}$	$\frac{\hbar\overline{y_{1,i_3,r_1}}\,\overline{L_{a_1,i_2}}\, P_R\, e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\hbar\overline{y_{1,i_3,r_1}}\,\overline{L_{a_1,i_2}}\, P_R\, e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}$ $+\frac{\hbar\overline{y_{2,i_3,r_1}}\,\overline{L_{a_1,i_2}}\, P_R\, e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\hbar\overline{y_{2,i_3,r_1}}\,\overline{L_{a_1,i_2}}\, P_R\, e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}$	$+\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3}\lambda_5\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}M_F}$ $+\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{\lambda_5 L_{a_3,i_1}}\, C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4} H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\sqrt{3}M_F}$
cont.								$+\frac{2\hbar\overline{\lambda_5 L_{a_3,i_1}}\, C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4} H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\sqrt{3}M_F}$	$+\frac{\hbar\overline{y_{1,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\, P_L\, L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}{}^\dagger}{4\epsilon}$ $+\frac{3}{8}\hbar\overline{y_{1,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\, P_L\, L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}{}^\dagger$ $+\frac{\hbar\overline{y_{2,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\, P_L\, L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}{}^\dagger}{4\epsilon}$	
cont.									$+\frac{3}{8}\hbar\overline{y_{2,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\, P_L\, L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}{}^\dagger$	
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3}\lambda_5\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}\epsilon M_F}$ $+\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3}\lambda_5\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}M_F}$ $+\frac{2\hbar\overline{\lambda_5 L_{a_3,i_1}}\, C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4} H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\sqrt{3}\epsilon M_F}$	$\frac{\hbar\overline{y_{1,i_3,r_1}}\,\overline{L_{a_1,i_2}}\, P_R\, e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\hbar\overline{y_{1,i_3,r_1}}\,\overline{L_{a_1,i_2}}\, P_R\, e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}$ $+\frac{\hbar\overline{y_{2,i_3,r_1}}\,\overline{L_{a_1,i_2}}\, P_R\, e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\hbar\overline{y_{2,i_3,r_1}}\,\overline{L_{a_1,i_2}}\, P_R\, e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}$	$+\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3}\lambda_5\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}M_F}$ $+\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{\lambda_5 L_{a_3,i_1}}\, C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4} H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\sqrt{3}M_F}$
cont.								$+\frac{2\hbar\overline{\lambda_5 L_{a_3,i_1}}\, C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4} H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\sqrt{3}M_F}$	$+\frac{\hbar\overline{y_{1,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\, P_L\, L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}{}^\dagger}{4\epsilon}$ $+\frac{3}{8}\hbar\overline{y_{1,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\, P_L\, L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}{}^\dagger$	
cont.									$+\frac{3}{8}\hbar\overline{y_{1,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\, P_L\, L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}{}^\dagger$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$H\,\ell$	$H\,\ell\,e$	$F\,H\,\ell$
cont.									$+\frac{\hbar \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\; P_L\; {L_{a_1,i_1}}\, y_{2,i_3,r_1}\, {H_{a_1}}^{\dagger}}{4\epsilon}$	
									$+\frac{3}{8}\hbar \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\; P_L\; {L_{a_1,i_1}}\, y_{2,i_3,r_1}\, {H_{a_1}}^{\dagger}$	
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+\frac{2\hbar \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} {L_{a_4,i_2}}^{T}\, C\, P_L\, L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_5 \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3}\epsilon M_F}$	$\frac{\hbar \overline{y_{1,i_3,r_1}} \overline{L_{a_1,i_2}}\; P_R\, e_{i_1} H_{a_1}\, y_{1,i_2,r_1}\, Y_{e,i_3,i_1}}{4\epsilon}$	$+\frac{2\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} {L_{a_4,i_2}}^{T}\, C\, P_L\, L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_5 \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3} M_F}$
								$+\frac{2\hbar \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} {L_{a_4,i_2}}^{T}\, C\, P_L\, L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_5 \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3} M_F}$	$+\frac{3}{8}\hbar \overline{y_{1,i_3,r_1}} \overline{L_{a_1,i_2}}\; P_R\, e_{i_1} H_{a_1}\, y_{1,i_2,r_1}\, Y_{e,i_3,i_1}$	
								$+\frac{2\hbar \overline{\lambda_5 L_{a_3,i_1}}\; C\, P_R\, \overline{{L_{a_4,i_2}}^{T}}\, y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} {H_{a_1}}^{\dagger} {H_{a_2}}^{\dagger}}{\sqrt{3}\epsilon M_F}$	$+\frac{\hbar \overline{y_{2,i_3,r_1}} \overline{L_{a_1,i_2}}\; P_R\, e_{i_1} H_{a_1}\, y_{2,i_2,r_1}\, Y_{e,i_3,i_1}}{4\epsilon}$	$+\frac{2\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{\lambda_5 L_{a_3,i_1}}\; C\, P_R\, \overline{{L_{a_4,i_2}}^{T}}\, y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} {H_{a_1}}^{\dagger} {H_{a_2}}^{\dagger}}{\sqrt{3} M_F}$
									$+\frac{3}{8}\hbar \overline{y_{2,i_3,r_1}} \overline{L_{a_1,i_2}}\; P_R\, e_{i_1} H_{a_1}\, y_{2,i_2,r_1}\, Y_{e,i_3,i_1}$	
cont.								$+\frac{2\hbar \overline{\lambda_5 L_{a_3,i_1}}\; C\, P_R\, \overline{{L_{a_4,i_2}}^{T}}\, y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} {H_{a_1}}^{\dagger} {H_{a_2}}^{\dagger}}{\sqrt{3} M_F}$	$+\frac{\hbar \overline{y_{1,i_1,r_1}} \overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\; P_L\; {L_{a_1,i_1}}\, y_{1,i_3,r_1}\, {H_{a_1}}^{\dagger}}{4\epsilon}$	
									$+\frac{3}{8}\hbar \overline{y_{1,i_1,r_1}} \overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\; P_L\; {L_{a_1,i_1}}\, y_{1,i_3,r_1}\, {H_{a_1}}^{\dagger}$	
									$+\frac{\hbar \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\; P_L\; {L_{a_1,i_1}}\, y_{2,i_3,r_1}\, {H_{a_1}}^{\dagger}}{4\epsilon}$	
cont.									$+\frac{3}{8}\hbar \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\; P_L\; {L_{a_1,i_1}}\, y_{2,i_3,r_1}\, {H_{a_1}}^{\dagger}$	
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$+\frac{2\hbar \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} {L_{a_4,i_2}}^{T}\, C\, P_L\, L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_5 \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3}\epsilon M_F}$	$\frac{\hbar \overline{y_{1,i_3,r_1}} \overline{L_{a_1,i_2}}\; P_R\, e_{i_1} H_{a_1}\, y_{1,i_2,r_1}\, Y_{e,i_3,i_1}}{4\epsilon}$	$+\frac{2\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} {L_{a_4,i_2}}^{T}\, C\, P_L\, L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_5 \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3} M_F}$
								$+\frac{2\hbar \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} {L_{a_4,i_2}}^{T}\, C\, P_L\, L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_5 \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3} M_F}$	$+\frac{3}{8}\hbar \overline{y_{1,i_3,r_1}} \overline{L_{a_1,i_2}}\; P_R\, e_{i_1} H_{a_1}\, y_{1,i_2,r_1}\, Y_{e,i_3,i_1}$	
								$+\frac{2\hbar \overline{\lambda_5 L_{a_3,i_1}}\; C\, P_R\, \overline{{L_{a_4,i_2}}^{T}}\, y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} {H_{a_1}}^{\dagger} {H_{a_2}}^{\dagger}}{\sqrt{3}\epsilon M_F}$	$+\frac{\hbar \overline{y_{2,i_3,r_1}} \overline{L_{a_1,i_2}}\; P_R\, e_{i_1} H_{a_1}\, y_{2,i_2,r_1}\, Y_{e,i_3,i_1}}{4\epsilon}$	$+\frac{2\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{\lambda_5 L_{a_3,i_1}}\; C\, P_R\, \overline{{L_{a_4,i_2}}^{T}}\, y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} {H_{a_1}}^{\dagger} {H_{a_2}}^{\dagger}}{\sqrt{3} M_F}$
									$+\frac{3}{8}\hbar \overline{y_{2,i_3,r_1}} \overline{L_{a_1,i_2}}\; P_R\, e_{i_1} H_{a_1}\, y_{2,i_2,r_1}\, Y_{e,i_3,i_1}$	
cont.								$+\frac{2\hbar \overline{\lambda_5 L_{a_3,i_1}}\; C\, P_R\, \overline{{L_{a_4,i_2}}^{T}}\, y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} {H_{a_1}}^{\dagger} {H_{a_2}}^{\dagger}}{\sqrt{3} M_F}$	$+\frac{\hbar \overline{y_{1,i_1,r_1}} \overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\; P_L\; {L_{a_1,i_1}}\, y_{1,i_3,r_1}\, {H_{a_1}}^{\dagger}}{4\epsilon}$	
									$+\frac{3}{8}\hbar \overline{y_{1,i_1,r_1}} \overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\; P_L\; {L_{a_1,i_1}}\, y_{1,i_3,r_1}\, {H_{a_1}}^{\dagger}$	
									$+\frac{\hbar \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\; P_L\; {L_{a_1,i_1}}\, y_{2,i_3,r_1}\, {H_{a_1}}^{\dagger}}{4\epsilon}$	
cont.									$+\frac{3}{8}\hbar \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\; P_L\; {L_{a_1,i_1}}\, y_{2,i_3,r_1}\, {H_{a_1}}^{\dagger}$	
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$+\frac{2\hbar \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} {L_{a_4,i_2}}^{T}\, C\, P_L\, L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_5 \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3}\epsilon M_F}$	$\frac{\hbar \overline{y_{1,i_3,r_1}} \overline{L_{a_1,i_2}}\; P_R\, e_{i_1} H_{a_1}\, y_{1,i_2,r_1}\, Y_{e,i_3,i_1}}{4\epsilon}$	$+\frac{2\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} {L_{a_4,i_2}}^{T}\, C\, P_L\, L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_5 \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3} M_F}$
								$+\frac{2\hbar \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} {L_{a_4,i_2}}^{T}\, C\, P_L\, L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_5 \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3} M_F}$	$+\frac{3}{8}\hbar \overline{y_{1,i_3,r_1}} \overline{L_{a_1,i_2}}\; P_R\, e_{i_1} H_{a_1}\, y_{1,i_2,r_1}\, Y_{e,i_3,i_1}$	
								$+\frac{2\hbar \overline{\lambda_5 L_{a_3,i_1}}\; C\, P_R\, \overline{{L_{a_4,i_2}}^{T}}\, y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} {H_{a_1}}^{\dagger} {H_{a_2}}^{\dagger}}{\sqrt{3}\epsilon M_F}$	$+\frac{\hbar \overline{y_{2,i_3,r_1}} \overline{L_{a_1,i_2}}\; P_R\, e_{i_1} H_{a_1}\, y_{2,i_2,r_1}\, Y_{e,i_3,i_1}}{4\epsilon}$	$+\frac{2\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{\lambda_5 L_{a_3,i_1}}\; C\, P_R\, \overline{{L_{a_4,i_2}}^{T}}\, y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} {H_{a_1}}^{\dagger} {H_{a_2}}^{\dagger}}{\sqrt{3} M_F}$
									$+\frac{3}{8}\hbar \overline{y_{2,i_3,r_1}} \overline{L_{a_1,i_2}}\; P_R\, e_{i_1} H_{a_1}\, y_{2,i_2,r_1}\, Y_{e,i_3,i_1}$	
cont.								$+\frac{2\hbar \overline{\lambda_5 L_{a_3,i_1}}\; C\, P_R\, \overline{{L_{a_4,i_2}}^{T}}\, y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} {H_{a_1}}^{\dagger} {H_{a_2}}^{\dagger}}{\sqrt{3} M_F}$	$+\frac{\hbar \overline{y_{1,i_1,r_1}} \overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\; P_L\; {L_{a_1,i_1}}\, y_{1,i_3,r_1}\, {H_{a_1}}^{\dagger}}{4\epsilon}$	
									$+\frac{3}{8}\hbar \overline{y_{1,i_1,r_1}} \overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\; P_L\; {L_{a_1,i_1}}\, y_{1,i_3,r_1}\, {H_{a_1}}^{\dagger}$	
									$+\frac{\hbar \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\; P_L\; {L_{a_1,i_1}}\, y_{2,i_3,r_1}\, {H_{a_1}}^{\dagger}}{4\epsilon}$	
cont.									$+\frac{3}{8}\hbar \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\; P_L\; {L_{a_1,i_1}}\, y_{2,i_3,r_1}\, {H_{a_1}}^{\dagger}$	
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$-\frac{2\hbar \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} {L_{a_4,i_2}}^{T}\, C\, P_L\, L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_5 \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{3\epsilon M_F}$	$\frac{\sqrt{3}\hbar \overline{y_{1,i_3,r_1}} \overline{L_{a_1,i_2}}\; P_R\, e_{i_1} H_{a_1}\, y_{1,i_2,r_1}\, Y_{e,i_3,i_1}}{4\epsilon}$	$-\frac{2\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} {L_{a_4,i_2}}^{T}\, C\, P_L\, L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_5 \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{3 M_F}$
								$-\frac{2\hbar \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} {L_{a_4,i_2}}^{T}\, C\, P_L\, L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_5 \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{3 M_F}$	$+\frac{3}{8}\sqrt{3}\hbar \overline{y_{1,i_3,r_1}} \overline{L_{a_1,i_2}}\; P_R\, e_{i_1} H_{a_1}\, y_{1,i_2,r_1}\, Y_{e,i_3,i_1}$	
								$-\frac{2\hbar \overline{\lambda_5 L_{a_3,i_1}}\; C\, P_R\, \overline{{L_{a_4,i_2}}^{T}}\, y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} {H_{a_1}}^{\dagger} {H_{a_2}}^{\dagger}}{3\epsilon M_F}$	$+\frac{\sqrt{3}\hbar \overline{y_{2,i_3,r_1}} \overline{L_{a_1,i_2}}\; P_R\, e_{i_1} H_{a_1}\, y_{2,i_2,r_1}\, Y_{e,i_3,i_1}}{4\epsilon}$	$-\frac{2\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{\lambda_5 L_{a_3,i_1}}\; C\, P_R\, \overline{{L_{a_4,i_2}}^{T}}\, y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} {H_{a_1}}^{\dagger} {H_{a_2}}^{\dagger}}{3 M_F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$H\,\ell$	$H\,\ell\,e$	$F\,H\,\ell$
								$-\frac{2\hbar\overline{\lambda_5 L_{a_3,i_1}}\,C\,P_R\,\overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3}\epsilon_{a_2,a_4} {H_{a_1}}^\dagger {H_{a_2}}^\dagger}{3M_F}$	$+\frac{3}{8}\sqrt{3}\hbar\overline{y_{2,i_3,r_1}}\,\overline{L_{a_1,i_2}}\,P_R\,e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}$ $+\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1} y_{1,i_3,r_1} {H_{a_1}}^\dagger}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1} y_{1,i_3,r_1} {H_{a_1}}^\dagger$ $+\frac{\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1} y_{2,i_3,r_1} {H_{a_1}}^\dagger}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1} y_{2,i_3,r_1} {H_{a_1}}^\dagger$	
cont.										
								$-\frac{2\hbar\overline{y_{1,i_1,r_1} y_{2,i_2,r_1}}\,{L_{a_4,i_2}}^T\,C\,P_L\,L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_5 \epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{3\epsilon M_F}$	$\frac{\sqrt{3}\hbar\overline{y_{1,i_3,r_1}}\,\overline{L_{a_1,i_2}}\,P_R\,e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$	$-\frac{2\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1} y_{2,i_2,r_1}}\,{L_{a_4,i_2}}^T\,C\,P_L\,L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_5 \epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{3M_F}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{2\hbar\overline{y_{1,i_1,r_1} y_{2,i_2,r_1}}\,{L_{a_4,i_2}}^T\,C\,P_L\,L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_5 \epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{3M_F}$	$+\frac{3}{8}\sqrt{3}\hbar\overline{y_{1,i_3,r_1}}\,\overline{L_{a_1,i_2}}\,P_R\,e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}$	$-\frac{2\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{\lambda_5 L_{a_3,i_1}}\,C\,P_R\,\overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3}\epsilon_{a_2,a_4} {H_{a_1}}^\dagger {H_{a_2}}^\dagger}{3M_F}$
								$-\frac{2\hbar\overline{\lambda_5 L_{a_3,i_1}}\,C\,P_R\,\overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3}\epsilon_{a_2,a_4} {H_{a_1}}^\dagger {H_{a_2}}^\dagger}{3\epsilon M_F}$	$+\frac{\sqrt{3}\hbar\overline{y_{2,i_3,r_1}}\,\overline{L_{a_1,i_2}}\,P_R\,e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$	
									$+\frac{3}{8}\sqrt{3}\hbar\overline{y_{2,i_3,r_1}}\,\overline{L_{a_1,i_2}}\,P_R\,e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}$	
cont.								$-\frac{2\hbar\overline{\lambda_5 L_{a_3,i_1}}\,C\,P_R\,\overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3}\epsilon_{a_2,a_4} {H_{a_1}}^\dagger {H_{a_2}}^\dagger}{3M_F}$	$+\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1} y_{1,i_3,r_1} {H_{a_1}}^\dagger}{4\epsilon}$	
									$+\frac{3}{8}\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1} y_{1,i_3,r_1} {H_{a_1}}^\dagger$	
									$+\frac{\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1} y_{2,i_3,r_1} {H_{a_1}}^\dagger}{4\epsilon}$	
cont.									$+\frac{3}{8}\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1} y_{2,i_3,r_1} {H_{a_1}}^\dagger$	
								$-\frac{2\hbar\overline{y_{1,i_1,r_1} y_{2,i_2,r_1}}\,{L_{a_4,i_2}}^T\,C\,P_L\,L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_5 \epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{3\epsilon M_F}$	$\frac{\sqrt{3}\hbar\overline{y_{1,i_3,r_1}}\,\overline{L_{a_1,i_2}}\,P_R\,e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$	$-\frac{2\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1} y_{2,i_2,r_1}}\,{L_{a_4,i_2}}^T\,C\,P_L\,L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_5 \epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{3M_F}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-\frac{2\hbar\overline{y_{1,i_1,r_1} y_{2,i_2,r_1}}\,{L_{a_4,i_2}}^T\,C\,P_L\,L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_5 \epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{3M_F}$	$+\frac{3}{8}\sqrt{3}\hbar\overline{y_{1,i_3,r_1}}\,\overline{L_{a_1,i_2}}\,P_R\,e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}$	$-\frac{2\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{\lambda_5 L_{a_3,i_1}}\,C\,P_R\,\overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3}\epsilon_{a_2,a_4} {H_{a_1}}^\dagger {H_{a_2}}^\dagger}{3M_F}$
								$-\frac{2\hbar\overline{\lambda_5 L_{a_3,i_1}}\,C\,P_R\,\overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3}\epsilon_{a_2,a_4} {H_{a_1}}^\dagger {H_{a_2}}^\dagger}{3\epsilon M_F}$	$+\frac{\sqrt{3}\hbar\overline{y_{2,i_3,r_1}}\,\overline{L_{a_1,i_2}}\,P_R\,e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$	
									$+\frac{3}{8}\sqrt{3}\hbar\overline{y_{2,i_3,r_1}}\,\overline{L_{a_1,i_2}}\,P_R\,e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}$	
cont.								$-\frac{2\hbar\overline{\lambda_5 L_{a_3,i_1}}\,C\,P_R\,\overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3}\epsilon_{a_2,a_4} {H_{a_1}}^\dagger {H_{a_2}}^\dagger}{3M_F}$	$+\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1} y_{1,i_3,r_1} {H_{a_1}}^\dagger}{4\epsilon}$	
									$+\frac{3}{8}\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1} y_{1,i_3,r_1} {H_{a_1}}^\dagger$	
									$+\frac{\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1} y_{2,i_3,r_1} {H_{a_1}}^\dagger}{4\epsilon}$	
cont.									$+\frac{3}{8}\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1} y_{2,i_3,r_1} {H_{a_1}}^\dagger$	
								$-\frac{2\hbar\overline{y_{1,i_1,r_1} y_{2,i_2,r_1}}\,{L_{a_4,i_2}}^T\,C\,P_L\,L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_5 \epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{3\epsilon M_F}$	$\frac{\sqrt{3}\hbar\overline{y_{1,i_3,r_1}}\,\overline{L_{a_1,i_2}}\,P_R\,e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$	$-\frac{2\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1} y_{2,i_2,r_1}}\,{L_{a_4,i_2}}^T\,C\,P_L\,L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_5 \epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{3M_F}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$-\frac{2\hbar\overline{y_{1,i_1,r_1} y_{2,i_2,r_1}}\,{L_{a_4,i_2}}^T\,C\,P_L\,L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_5 \epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{3M_F}$	$+\frac{3}{8}\sqrt{3}\hbar\overline{y_{1,i_3,r_1}}\,\overline{L_{a_1,i_2}}\,P_R\,e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}$	$-\frac{2\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{\lambda_5 L_{a_3,i_1}}\,C\,P_R\,\overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3}\epsilon_{a_2,a_4} {H_{a_1}}^\dagger {H_{a_2}}^\dagger}{3M_F}$
								$-\frac{2\hbar\overline{\lambda_5 L_{a_3,i_1}}\,C\,P_R\,\overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3}\epsilon_{a_2,a_4} {H_{a_1}}^\dagger {H_{a_2}}^\dagger}{3\epsilon M_F}$	$+\frac{\sqrt{3}\hbar\overline{y_{2,i_3,r_1}}\,\overline{L_{a_1,i_2}}\,P_R\,e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$	
									$+\frac{3}{8}\sqrt{3}\hbar\overline{y_{2,i_3,r_1}}\,\overline{L_{a_1,i_2}}\,P_R\,e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}$	
cont.								$-\frac{2\hbar\overline{\lambda_5 L_{a_3,i_1}}\,C\,P_R\,\overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3}\epsilon_{a_2,a_4} {H_{a_1}}^\dagger {H_{a_2}}^\dagger}{3M_F}$	$+\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1} y_{1,i_3,r_1} {H_{a_1}}^\dagger}{4\epsilon}$	
									$+\frac{3}{8}\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1} y_{1,i_3,r_1} {H_{a_1}}^\dagger$	
									$+\frac{\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1} y_{2,i_3,r_1} {H_{a_1}}^\dagger}{4\epsilon}$	
cont.									$+\frac{3}{8}\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1} y_{2,i_3,r_1} {H_{a_1}}^\dagger$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$H\,\ell$	$H\,\ell\,e$	$F\,H\,\ell$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$-\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}{L_{a_4,i_2}}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_5 \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{3\epsilon M_F}$ $-\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}{L_{a_4,i_2}}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_5 \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{3M_F}$ $-\frac{2\hbar\overline{\lambda_5 L_{a_3,i_1}}\, C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} {H_{a_1}}^\dagger {H_{a_2}}^\dagger}{3\epsilon M_F}$	$\frac{\sqrt{3}\hbar\overline{y_{1,i_3,r_1}}\,\overline{L_{a_1,i_2}}\, P_R\, e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\hbar\overline{y_{1,i_3,r_1}}\,\overline{L_{a_1,i_2}}\, P_R\, e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}$ $+\frac{\sqrt{3}\hbar\overline{y_{2,i_3,r_1}}\,\overline{L_{a_1,i_2}}\, P_R\, e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\hbar\overline{y_{2,i_3,r_1}}\,\overline{L_{a_1,i_2}}\, P_R\, e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}$ $+\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\, P_L\, L_{a_1,i_1} y_{1,i_3,r_1} {H_{a_1}}^\dagger}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\, P_L\, L_{a_1,i_1} y_{1,i_3,r_1} {H_{a_1}}^\dagger$ $+\frac{\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\, P_L\, L_{a_1,i_1} y_{2,i_3,r_1} {H_{a_1}}^\dagger}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\, P_L\, L_{a_1,i_1} y_{2,i_3,r_1} {H_{a_1}}^\dagger$	$-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}{L_{a_4,i_2}}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_5 \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{3M_F}$ $-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{\lambda_5 L_{a_3,i_1}}\, C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} {H_{a_1}}^\dagger {H_{a_2}}^\dagger}{3M_F}$
cont.										
cont.										
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}{L_{a_4,i_2}}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_5 \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3}\epsilon M_F}$ $+\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}{L_{a_4,i_2}}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_5 \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3}M_F}$ $+\frac{2\hbar\overline{\lambda_5 L_{a_3,i_1}}\, C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} {H_{a_1}}^\dagger {H_{a_2}}^\dagger}{\sqrt{3}\epsilon M_F}$	$\frac{\sqrt{3}\hbar\overline{y_{1,i_3,r_1}}\,\overline{L_{a_1,i_2}}\, P_R\, e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\hbar\overline{y_{1,i_3,r_1}}\,\overline{L_{a_1,i_2}}\, P_R\, e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}$ $+\frac{\hbar\overline{y_{2,i_3,r_1}}\,\overline{L_{a_1,i_2}}\, P_R\, e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\hbar\overline{y_{2,i_3,r_1}}\,\overline{L_{a_1,i_2}}\, P_R\, e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}$ $+\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\, P_L\, L_{a_1,i_1} y_{1,i_3,r_1} {H_{a_1}}^\dagger}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\, P_L\, L_{a_1,i_1} y_{1,i_3,r_1} {H_{a_1}}^\dagger$ $+\frac{\hbar\overline{y_{2,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\, P_L\, L_{a_1,i_1} y_{2,i_3,r_1} {H_{a_1}}^\dagger}{4\epsilon}$ $+\frac{3}{8}\hbar\overline{y_{2,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\, P_L\, L_{a_1,i_1} y_{2,i_3,r_1} {H_{a_1}}^\dagger$	$+\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}{L_{a_4,i_2}}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_5 \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3}M_F}$ $+\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{\lambda_5 L_{a_3,i_1}}\, C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} {H_{a_1}}^\dagger {H_{a_2}}^\dagger}{\sqrt{3}M_F}$
cont.										
cont.										
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$+\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}{L_{a_4,i_2}}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_5 \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3}\epsilon M_F}$ $+\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}{L_{a_4,i_2}}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_5 \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3}M_F}$ $+\frac{2\hbar\overline{\lambda_5 L_{a_3,i_1}}\, C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} {H_{a_1}}^\dagger {H_{a_2}}^\dagger}{\sqrt{3}\epsilon M_F}$	$\frac{\sqrt{3}\hbar\overline{y_{1,i_3,r_1}}\,\overline{L_{a_1,i_2}}\, P_R\, e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\hbar\overline{y_{1,i_3,r_1}}\,\overline{L_{a_1,i_2}}\, P_R\, e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}$ $+\frac{\hbar\overline{y_{2,i_3,r_1}}\,\overline{L_{a_1,i_2}}\, P_R\, e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\hbar\overline{y_{2,i_3,r_1}}\,\overline{L_{a_1,i_2}}\, P_R\, e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}$ $+\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\, P_L\, L_{a_1,i_1} y_{1,i_3,r_1} {H_{a_1}}^\dagger}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\, P_L\, L_{a_1,i_1} y_{1,i_3,r_1} {H_{a_1}}^\dagger$ $+\frac{\hbar\overline{y_{2,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\, P_L\, L_{a_1,i_1} y_{2,i_3,r_1} {H_{a_1}}^\dagger}{4\epsilon}$ $+\frac{3}{8}\hbar\overline{y_{2,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\, P_L\, L_{a_1,i_1} y_{2,i_3,r_1} {H_{a_1}}^\dagger$	$+\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}{L_{a_4,i_2}}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_5 \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3}M_F}$ $+\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{\lambda_5 L_{a_3,i_1}}\, C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} {H_{a_1}}^\dagger {H_{a_2}}^\dagger}{\sqrt{3}M_F}$
cont.										
cont.										
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}{L_{a_4,i_2}}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_5 \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3}\epsilon M_F}$ $+\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}{L_{a_4,i_2}}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_5 \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3}M_F}$ $+\frac{2\hbar\overline{\lambda_5 L_{a_3,i_1}}\, C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} {H_{a_1}}^\dagger {H_{a_2}}^\dagger}{\sqrt{3}\epsilon M_F}$	$\frac{\sqrt{3}\hbar\overline{y_{1,i_3,r_1}}\,\overline{L_{a_1,i_2}}\, P_R\, e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\hbar\overline{y_{1,i_3,r_1}}\,\overline{L_{a_1,i_2}}\, P_R\, e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}$ $+\frac{\hbar\overline{y_{2,i_3,r_1}}\,\overline{L_{a_1,i_2}}\, P_R\, e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\hbar\overline{y_{2,i_3,r_1}}\,\overline{L_{a_1,i_2}}\, P_R\, e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}$ $+\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\, P_L\, L_{a_1,i_1} y_{1,i_3,r_1} {H_{a_1}}^\dagger}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\, P_L\, L_{a_1,i_1} y_{1,i_3,r_1} {H_{a_1}}^\dagger$	$+\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}{L_{a_4,i_2}}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_5 \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3}M_F}$ $+\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{\lambda_5 L_{a_3,i_1}}\, C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} {H_{a_1}}^\dagger {H_{a_2}}^\dagger}{\sqrt{3}M_F}$
cont.										

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$H\,\ell$	$H\,\ell\,e$	$F\,H\,\ell$
cont.									$+\frac{\hbar \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}}\, P_L\, L_{a_1,i_1} y_{2,i_3,r_1} {H_{a_1}}^\dagger}{4\epsilon}$ $+\frac{3}{8}\hbar \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}}\, P_L\, L_{a_1,i_1} y_{2,i_3,r_1} {H_{a_1}}^\dagger$	
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$+\frac{2\hbar \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} {L_{a_4,i_2}}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_5 \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3}\epsilon M_F}$ $+\frac{2\hbar \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} {L_{a_4,i_2}}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_5 \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3} M_F}$ $+\frac{2\hbar \overline{\lambda_5 L_{a_3,i_1}}\, C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} {H_{a_1}}^\dagger {H_{a_2}}^\dagger}{\sqrt{3}\epsilon M_F}$	$\frac{\sqrt{3}\hbar \overline{y_{1,i_3,r_1}} \overline{L_{a_1,i_2}}\, P_R\, e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\hbar \overline{y_{1,i_3,r_1}} \overline{L_{a_1,i_2}}\, P_R\, e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}$ $+\frac{\hbar \overline{y_{2,i_3,r_1}} \overline{L_{a_1,i_2}}\, P_R\, e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\hbar \overline{y_{2,i_3,r_1}} \overline{L_{a_1,i_2}}\, P_R\, e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}$	$+\frac{2\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} {L_{a_4,i_2}}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_5 \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3} M_F}$ $+\frac{2\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{\lambda_5 L_{a_3,i_1}}\, C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} {H_{a_1}}^\dagger {H_{a_2}}^\dagger}{\sqrt{3} M_F}$
cont.								$+\frac{2\hbar \overline{\lambda_5 L_{a_3,i_1}}\, C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} {H_{a_1}}^\dagger {H_{a_2}}^\dagger}{\sqrt{3} M_F}$	$+\frac{\sqrt{3}\hbar \overline{y_{1,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}}\, P_L\, L_{a_1,i_1} y_{1,i_3,r_1} {H_{a_1}}^\dagger}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\hbar \overline{y_{1,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}}\, P_L\, L_{a_1,i_1} y_{1,i_3,r_1} {H_{a_1}}^\dagger$	
cont.									$+\frac{\hbar \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}}\, P_L\, L_{a_1,i_1} y_{2,i_3,r_1} {H_{a_1}}^\dagger}{4\epsilon}$ $+\frac{3}{8}\hbar \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}}\, P_L\, L_{a_1,i_1} y_{2,i_3,r_1} {H_{a_1}}^\dagger$	
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$+\frac{2\hbar \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} {L_{a_4,i_2}}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_5 \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3}\epsilon M_F}$ $+\frac{2\hbar \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} {L_{a_4,i_2}}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_5 \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3} M_F}$ $+\frac{2\hbar \overline{\lambda_5 L_{a_3,i_1}}\, C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} {H_{a_1}}^\dagger {H_{a_2}}^\dagger}{\sqrt{3}\epsilon M_F}$	$\frac{\sqrt{3}\hbar \overline{y_{1,i_3,r_1}} \overline{L_{a_1,i_2}}\, P_R\, e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\hbar \overline{y_{1,i_3,r_1}} \overline{L_{a_1,i_2}}\, P_R\, e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}$ $+\frac{\hbar \overline{y_{2,i_3,r_1}} \overline{L_{a_1,i_2}}\, P_R\, e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\hbar \overline{y_{2,i_3,r_1}} \overline{L_{a_1,i_2}}\, P_R\, e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}$	$+\frac{2\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} {L_{a_4,i_2}}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_5 \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3} M_F}$ $+\frac{2\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{\lambda_5 L_{a_3,i_1}}\, C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} {H_{a_1}}^\dagger {H_{a_2}}^\dagger}{\sqrt{3} M_F}$
cont.								$+\frac{2\hbar \overline{\lambda_5 L_{a_3,i_1}}\, C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} {H_{a_1}}^\dagger {H_{a_2}}^\dagger}{\sqrt{3} M_F}$	$+\frac{\sqrt{3}\hbar \overline{y_{1,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}}\, P_L\, L_{a_1,i_1} y_{1,i_3,r_1} {H_{a_1}}^\dagger}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\hbar \overline{y_{1,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}}\, P_L\, L_{a_1,i_1} y_{1,i_3,r_1} {H_{a_1}}^\dagger$	
cont.									$+\frac{\hbar \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}}\, P_L\, L_{a_1,i_1} y_{2,i_3,r_1} {H_{a_1}}^\dagger}{4\epsilon}$ $+\frac{3}{8}\hbar \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}}\, P_L\, L_{a_1,i_1} y_{2,i_3,r_1} {H_{a_1}}^\dagger$	
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$-\frac{2\sqrt{\frac{2}{3}}\hbar \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} {L_{a_4,i_2}}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_5 \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\epsilon M_F}$ $-\frac{2\sqrt{\frac{2}{3}}\hbar \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} {L_{a_4,i_2}}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_5 \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{M_F}$ $-\frac{2\sqrt{\frac{2}{3}}\hbar \overline{\lambda_5 L_{a_3,i_1}}\, C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} {H_{a_1}}^\dagger {H_{a_2}}^\dagger}{\epsilon M_F}$	$\frac{\sqrt{3}\hbar \overline{y_{1,i_3,r_1}} \overline{L_{a_1,i_2}}\, P_R\, e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\hbar \overline{y_{1,i_3,r_1}} \overline{L_{a_1,i_2}}\, P_R\, e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}$ $+\frac{\sqrt{3}\hbar \overline{y_{2,i_3,r_1}} \overline{L_{a_1,i_2}}\, P_R\, e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\hbar \overline{y_{2,i_3,r_1}} \overline{L_{a_1,i_2}}\, P_R\, e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}$	$-\frac{2\sqrt{\frac{2}{3}}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} {L_{a_4,i_2}}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_5 \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{M_F}$ $-\frac{2\sqrt{\frac{2}{3}}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{\lambda_5 L_{a_3,i_1}}\, C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} {H_{a_1}}^\dagger {H_{a_2}}^\dagger}{M_F}$
cont.								$-\frac{2\sqrt{\frac{2}{3}}\hbar \overline{\lambda_5 L_{a_3,i_1}}\, C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} {H_{a_1}}^\dagger {H_{a_2}}^\dagger}{M_F}$	$+\frac{\sqrt{3}\hbar \overline{y_{1,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}}\, P_L\, L_{a_1,i_1} y_{1,i_3,r_1} {H_{a_1}}^\dagger}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\hbar \overline{y_{1,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}}\, P_L\, L_{a_1,i_1} y_{1,i_3,r_1} {H_{a_1}}^\dagger$	
cont.									$+\frac{\sqrt{3}\hbar \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}}\, P_L\, L_{a_1,i_1} y_{2,i_3,r_1} {H_{a_1}}^\dagger}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\hbar \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_3,i_2}} \overline{e_{i_2}}\, P_L\, L_{a_1,i_1} y_{2,i_3,r_1} {H_{a_1}}^\dagger$	
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-\frac{2\sqrt{\frac{2}{3}}\hbar \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} {L_{a_4,i_2}}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_5 \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\epsilon M_F}$ $-\frac{2\sqrt{\frac{2}{3}}\hbar \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} {L_{a_4,i_2}}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_5 \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{M_F}$ $-\frac{2\sqrt{\frac{2}{3}}\hbar \overline{\lambda_5 L_{a_3,i_1}}\, C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} {H_{a_1}}^\dagger {H_{a_2}}^\dagger}{\epsilon M_F}$	$\frac{\sqrt{3}\hbar \overline{y_{1,i_3,r_1}} \overline{L_{a_1,i_2}}\, P_R\, e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\hbar \overline{y_{1,i_3,r_1}} \overline{L_{a_1,i_2}}\, P_R\, e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}$ $+\frac{\sqrt{3}\hbar \overline{y_{2,i_3,r_1}} \overline{L_{a_1,i_2}}\, P_R\, e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$	$-\frac{2\sqrt{\frac{2}{3}}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} {L_{a_4,i_2}}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_5 \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{M_F}$ $-\frac{2\sqrt{\frac{2}{3}}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{\lambda_5 L_{a_3,i_1}}\, C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} {H_{a_1}}^\dagger {H_{a_2}}^\dagger}{M_F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$H\,\ell$	$H\,\ell\,e$	$F\,H\,\ell$
								$-\frac{2\sqrt{\frac{2}{3}}\hbar\overline{\lambda_5 L_{a_3,i_1}}\,C\,P_R\,\overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4} H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{M_F}$	$+\frac{3}{8}\sqrt{3}\hbar\overline{y_{2,i_3,r_1}}\,\overline{L_{a_1,i_2}}\,P_R\,e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}$	
									$+\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}{}^\dagger}{4\epsilon}$	
									$+\frac{3}{8}\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}{}^\dagger$	
cont.									$+\frac{\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}{}^\dagger}{4\epsilon}$	
cont.									$+\frac{3}{8}\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}{}^\dagger$	
								$-\frac{2\sqrt{\frac{2}{3}}\hbar\overline{y_{1,i_1,r_1}\,y_{2,i_2,r_1}}L_{a_4,i_2}{}^T\,C\,P_L\,L_{a_2,i_1} H_{a_1} H_{a_3}\lambda_5\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\epsilon M_F}$	$\frac{\sqrt{3}\hbar\overline{y_{1,i_3,r_1}}\,\overline{L_{a_1,i_2}}\,P_R\,e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$	$-\frac{2\sqrt{\frac{2}{3}}\hbar\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}\,y_{2,i_2,r_1}}L_{a_4,i_2}{}^T\,C\,P_L\,L_{a_2,i_1} H_{a_1} H_{a_3}\lambda_5\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{M_F}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{2\sqrt{\frac{2}{3}}\hbar\overline{y_{1,i_1,r_1}\,y_{2,i_2,r_1}}L_{a_4,i_2}{}^T\,C\,P_L\,L_{a_2,i_1} H_{a_1} H_{a_3}\lambda_5\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{M_F}$	$+\frac{3}{8}\sqrt{3}\hbar\overline{y_{1,i_3,r_1}}\,\overline{L_{a_1,i_2}}\,P_R\,e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}$	$-\frac{2\sqrt{\frac{2}{3}}\hbar\log\left(\frac{\mu^2}{M_F^2}\right)\overline{\lambda_5 L_{a_3,i_1}}\,C\,P_R\,\overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4} H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{M_F}$
								$-\frac{2\sqrt{\frac{2}{3}}\hbar\overline{\lambda_5 L_{a_3,i_1}}\,C\,P_R\,\overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4} H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\epsilon M_F}$	$+\frac{\sqrt{3}\hbar\overline{y_{2,i_3,r_1}}\,\overline{L_{a_1,i_2}}\,P_R\,e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$	
									$+\frac{3}{8}\sqrt{3}\hbar\overline{y_{2,i_3,r_1}}\,\overline{L_{a_1,i_2}}\,P_R\,e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}$	
cont.									$+\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}{}^\dagger}{4\epsilon}$	
									$+\frac{3}{8}\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}{}^\dagger$	
									$+\frac{\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}{}^\dagger}{4\epsilon}$	
cont.									$+\frac{3}{8}\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}{}^\dagger$	
								$-\frac{2\sqrt{\frac{2}{3}}\hbar\overline{y_{1,i_1,r_1}\,y_{2,i_2,r_1}}L_{a_4,i_2}{}^T\,C\,P_L\,L_{a_2,i_1} H_{a_1} H_{a_3}\lambda_5\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\epsilon M_F}$	$\frac{\sqrt{3}\hbar\overline{y_{1,i_3,r_1}}\,\overline{L_{a_1,i_2}}\,P_R\,e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$	$-\frac{2\sqrt{\frac{2}{3}}\hbar\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}\,y_{2,i_2,r_1}}L_{a_4,i_2}{}^T\,C\,P_L\,L_{a_2,i_1} H_{a_1} H_{a_3}\lambda_5\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{M_F}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$-\frac{2\sqrt{\frac{2}{3}}\hbar\overline{y_{1,i_1,r_1}\,y_{2,i_2,r_1}}L_{a_4,i_2}{}^T\,C\,P_L\,L_{a_2,i_1} H_{a_1} H_{a_3}\lambda_5\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{M_F}$	$+\frac{3}{8}\sqrt{3}\hbar\overline{y_{1,i_3,r_1}}\,\overline{L_{a_1,i_2}}\,P_R\,e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}$	$-\frac{2\sqrt{\frac{2}{3}}\hbar\log\left(\frac{\mu^2}{M_F^2}\right)\overline{\lambda_5 L_{a_3,i_1}}\,C\,P_R\,\overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4} H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{M_F}$
								$-\frac{2\sqrt{\frac{2}{3}}\hbar\overline{\lambda_5 L_{a_3,i_1}}\,C\,P_R\,\overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4} H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\epsilon M_F}$	$+\frac{\sqrt{3}\hbar\overline{y_{2,i_3,r_1}}\,\overline{L_{a_1,i_2}}\,P_R\,e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$	
									$+\frac{3}{8}\sqrt{3}\hbar\overline{y_{2,i_3,r_1}}\,\overline{L_{a_1,i_2}}\,P_R\,e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}$	
cont.									$+\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}{}^\dagger}{4\epsilon}$	
									$+\frac{3}{8}\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}{}^\dagger$	
									$+\frac{\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}{}^\dagger}{4\epsilon}$	
cont.									$+\frac{3}{8}\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}{}^\dagger$	
								$-\frac{2\sqrt{\frac{2}{3}}\hbar\overline{y_{1,i_1,r_1}\,y_{2,i_2,r_1}}L_{a_4,i_2}{}^T\,C\,P_L\,L_{a_2,i_1} H_{a_1} H_{a_3}\lambda_5\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\epsilon M_F}$	$\frac{\sqrt{3}\hbar\overline{y_{1,i_3,r_1}}\,\overline{L_{a_1,i_2}}\,P_R\,e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$	$-\frac{2\sqrt{\frac{2}{3}}\hbar\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}\,y_{2,i_2,r_1}}L_{a_4,i_2}{}^T\,C\,P_L\,L_{a_2,i_1} H_{a_1} H_{a_3}\lambda_5\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{M_F}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$-\frac{2\sqrt{\frac{2}{3}}\hbar\overline{y_{1,i_1,r_1}\,y_{2,i_2,r_1}}L_{a_4,i_2}{}^T\,C\,P_L\,L_{a_2,i_1} H_{a_1} H_{a_3}\lambda_5\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{M_F}$	$+\frac{3}{8}\sqrt{3}\hbar\overline{y_{1,i_3,r_1}}\,\overline{L_{a_1,i_2}}\,P_R\,e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}$	$-\frac{2\sqrt{\frac{2}{3}}\hbar\log\left(\frac{\mu^2}{M_F^2}\right)\overline{\lambda_5 L_{a_3,i_1}}\,C\,P_R\,\overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4} H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{M_F}$
								$-\frac{2\sqrt{\frac{2}{3}}\hbar\overline{\lambda_5 L_{a_3,i_1}}\,C\,P_R\,\overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4} H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\epsilon M_F}$	$+\frac{\sqrt{3}\hbar\overline{y_{2,i_3,r_1}}\,\overline{L_{a_1,i_2}}\,P_R\,e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$	
									$+\frac{3}{8}\sqrt{3}\hbar\overline{y_{2,i_3,r_1}}\,\overline{L_{a_1,i_2}}\,P_R\,e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}$	
cont.									$+\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}{}^\dagger}{4\epsilon}$	
									$+\frac{3}{8}\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}{}^\dagger$	
									$+\frac{\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}{}^\dagger}{4\epsilon}$	
cont.									$+\frac{3}{8}\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}{}^\dagger$	

Field combinations 2 of 2

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$F\ H\ \ell\ e$
								$+\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}{H_{a_1}}^\dagger$
cont.								$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}{H_{a_1}}^\dagger$
								$+\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}{H_{a_1}}^\dagger$
cont.								$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}{H_{a_1}}^\dagger$
								$+\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}{H_{a_1}}^\dagger$
cont.								$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}{H_{a_1}}^\dagger$
								$+\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}{H_{a_1}}^\dagger$
cont.								$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}{H_{a_1}}^\dagger$
								$+\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}{H_{a_1}}^\dagger$
cont.								$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}{H_{a_1}}^\dagger$
								$+\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$+\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}{H_{a_1}}^\dagger$
cont.								$+\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}{H_{a_1}}^\dagger$
								$+\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}{H_{a_1}}^\dagger$
cont.								$+\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}{H_{a_1}}^\dagger$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$F\ H\ \ell\ e$
								$+\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}\overline{L_{a_1,i_2}}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}\overline{L_{a_1,i_2}}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}\overline{Y_{e,i_3,i_2}\overline{e_{i_2}}}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}{H_{a_1}}^\dagger$
cont.								$+\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}\overline{Y_{e,i_3,i_2}\overline{e_{i_2}}}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}{H_{a_1}}^\dagger$
								$+\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}\overline{L_{a_1,i_2}}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$+\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}\overline{L_{a_1,i_2}}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}\overline{Y_{e,i_3,i_2}\overline{e_{i_2}}}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}{H_{a_1}}^\dagger$
cont.								$+\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}\overline{Y_{e,i_3,i_2}\overline{e_{i_2}}}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}{H_{a_1}}^\dagger$
								$+\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}\overline{L_{a_1,i_2}}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$+\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}\overline{L_{a_1,i_2}}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}\overline{Y_{e,i_3,i_2}\overline{e_{i_2}}}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}{H_{a_1}}^\dagger$
cont.								$+\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}\overline{Y_{e,i_3,i_2}\overline{e_{i_2}}}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}{H_{a_1}}^\dagger$
								$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}\overline{L_{a_1,i_2}}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}\overline{L_{a_1,i_2}}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}\overline{Y_{e,i_3,i_2}\overline{e_{i_2}}}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}{H_{a_1}}^\dagger$
cont.								$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}\overline{Y_{e,i_3,i_2}\overline{e_{i_2}}}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}{H_{a_1}}^\dagger$
								$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}\overline{L_{a_1,i_2}}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}\overline{L_{a_1,i_2}}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}\overline{Y_{e,i_3,i_2}\overline{e_{i_2}}}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}{H_{a_1}}^\dagger$
cont.								$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}\overline{Y_{e,i_3,i_2}\overline{e_{i_2}}}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}{H_{a_1}}^\dagger$
								$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}\overline{L_{a_1,i_2}}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}\overline{L_{a_1,i_2}}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}\overline{Y_{e,i_3,i_2}\overline{e_{i_2}}}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}{H_{a_1}}^\dagger$
cont.								$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}\overline{Y_{e,i_3,i_2}\overline{e_{i_2}}}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}{H_{a_1}}^\dagger$
								$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}\overline{L_{a_1,i_2}}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}\overline{L_{a_1,i_2}}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}\overline{Y_{e,i_3,i_2}\overline{e_{i_2}}}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}{H_{a_1}}^\dagger$
cont.								$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}\overline{Y_{e,i_3,i_2}\overline{e_{i_2}}}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}{H_{a_1}}^\dagger$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$F\ H\ \ell\ e$
								$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}\overline{L_{a_1,i_2}}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}\overline{L_{a_1,i_2}}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}\overline{Y_{e,i_3,i_2}\overline{e_{i_2}}}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}{H_{a_1}}^\dagger$
cont.								$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}\overline{Y_{e,i_3,i_2}\overline{e_{i_2}}}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}{H_{a_1}}^\dagger$
								$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}\overline{L_{a_1,i_2}}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}\overline{L_{a_1,i_2}}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}\overline{Y_{e,i_3,i_2}\overline{e_{i_2}}}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}{H_{a_1}}^\dagger$
cont.								$+\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}\overline{Y_{e,i_3,i_2}\overline{e_{i_2}}}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}{H_{a_1}}^\dagger$
								$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}\overline{L_{a_1,i_2}}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$+\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}\overline{L_{a_1,i_2}}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}\overline{Y_{e,i_3,i_2}\overline{e_{i_2}}}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}{H_{a_1}}^\dagger$
cont.								$+\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}\overline{Y_{e,i_3,i_2}\overline{e_{i_2}}}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}{H_{a_1}}^\dagger$
								$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}\overline{L_{a_1,i_2}}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}\overline{L_{a_1,i_2}}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}\overline{Y_{e,i_3,i_2}\overline{e_{i_2}}}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}{H_{a_1}}^\dagger$
cont.								$+\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}\overline{Y_{e,i_3,i_2}\overline{e_{i_2}}}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}{H_{a_1}}^\dagger$
								$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}\overline{L_{a_1,i_2}}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$+\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}\overline{L_{a_1,i_2}}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}\overline{Y_{e,i_3,i_2}\overline{e_{i_2}}}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}{H_{a_1}}^\dagger$
cont.								$+\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}\overline{Y_{e,i_3,i_2}\overline{e_{i_2}}}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}{H_{a_1}}^\dagger$
								$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}\overline{L_{a_1,i_2}}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$+\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}\overline{L_{a_1,i_2}}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}\overline{Y_{e,i_3,i_2}\overline{e_{i_2}}}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}{H_{a_1}}^\dagger$
cont.								$+\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}\overline{Y_{e,i_3,i_2}\overline{e_{i_2}}}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}{H_{a_1}}^\dagger$
								$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}\overline{L_{a_1,i_2}}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}\overline{L_{a_1,i_2}}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}\overline{Y_{e,i_3,i_2}\overline{e_{i_2}}}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}{H_{a_1}}^\dagger$
cont.								$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}\overline{Y_{e,i_3,i_2}\overline{e_{i_2}}}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}{H_{a_1}}^\dagger$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$F\ H\ \ell\ e$
								$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}}\,\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}\,H_{a_1}\,y_{1,i_2,r_1}\,Y_{e,i_3,i_1}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}}\,\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}\,H_{a_1}\,y_{2,i_2,r_1}\,Y_{e,i_3,i_1}$
								$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}\,y_{1,i_3,r_1}\,{H_{a_1}}^\dagger$
cont.								$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}\,y_{2,i_3,r_1}\,{H_{a_1}}^\dagger$
								$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}}\,\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}\,H_{a_1}\,y_{1,i_2,r_1}\,Y_{e,i_3,i_1}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}}\,\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}\,H_{a_1}\,y_{2,i_2,r_1}\,Y_{e,i_3,i_1}$
								$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}\,y_{1,i_3,r_1}\,{H_{a_1}}^\dagger$
cont.								$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}\,y_{2,i_3,r_1}\,{H_{a_1}}^\dagger$
								$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}}\,\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}\,H_{a_1}\,y_{1,i_2,r_1}\,Y_{e,i_3,i_1}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}}\,\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}\,H_{a_1}\,y_{2,i_2,r_1}\,Y_{e,i_3,i_1}$
								$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}\,y_{1,i_3,r_1}\,{H_{a_1}}^\dagger$
cont.								$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}\,y_{2,i_3,r_1}\,{H_{a_1}}^\dagger$
								$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}}\,\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}\,H_{a_1}\,y_{1,i_2,r_1}\,Y_{e,i_3,i_1}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}}\,\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}\,H_{a_1}\,y_{2,i_2,r_1}\,Y_{e,i_3,i_1}$
								$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}\,y_{1,i_3,r_1}\,{H_{a_1}}^\dagger$
cont.								$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}\,y_{2,i_3,r_1}\,{H_{a_1}}^\dagger$